

Admissibility & Authority at the Building ML Consequence Boundary

A practitioner reference for building analytics, machine learning, controls, and autonomous execution.

ADMISSIBLE TO WHAT?	AUTHORITY TO DO WHAT?
The specific decision path that may produce consequence. Is this evidence fit to support <i>this</i> proposed action, under the rules and conditions that apply now?	Cause that specific consequence. Does this person, system, role, or institution have the established permission to make <i>this</i> action real now?

Three distinctions that prevent semantic collapse

VALID	ADMISSIBLE	AUTHORIZED
Is the observation or computation technically sound? Example: the sensor reading is accurate.	May it legitimately support this proposed decision? Validity alone does not answer this.	May this actor or system cause the proposed consequence? Admissible evidence alone does not answer this.

The governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Not just courtroom admissibility. Legal admissibility can be one downstream context. TA-14 asks an earlier architectural question: may this evidence legitimately support this proposed action now? A governed record may later need to survive audit, regulatory, contractual, dispute, or legal scrutiny.

The same inference can propose very different consequences

Machine learning can reconstruct missing data, predict degradation, classify faults, and forecast future performance. That capability is valuable. But the governance requirement attaches to the **proposed consequence**, not merely to the model.

SAME ML INFERENCE	PROPOSED CONSEQUENCE	WHY THE BOUNDARY CHANGES
AHU degradation predicted	1. Display an advisory	Low direct physical consequence. Evidence still needs provenance and context, but execution authority may be narrow.
AHU degradation predicted	2. Create a work order	Now the proposition affects workflow, labor, priority, and possibly contractual obligations.
AHU degradation predicted	3. Change a setpoint	The proposition crosses into control. Equipment and occupied-space conditions may change.
AHU degradation predicted	4. Shut equipment down	Potentially material consequence. Safety, occupancy, process, owner, code, policy, and operational authority may all matter.

Same model. Same inference. Different proposed consequence.
The evidence and authority sufficient to display an advisory should not automatically be assumed sufficient to alter a setpoint or shut down equipment.

Three rules for building ML

MODEL CONFIDENCE	DETERMINISM	PREDICTION
Model confidence != evidentiary sufficiency. A high-confidence inference can still lack required context, continuity, corroboration, or applicability.	Determinism != authority. A repeatable result can still propose an action the system is not permitted to cause.	Prediction != permission. Forecasting what is likely to happen does not establish what may be done about it.

An admissible input does not create an admissible output. Even when ML receives governed evidence, its recommendation is a new proposition: new proposition, new consequence, new NOW.

The execution boundary

The critical control surface is the last governed point before a proposition becomes a real-world effect.

KNOW	INFER	RECOMMEND	COMMIT BOUNDARY	ACT	VERIFY
Reality + record	Computation	Proposed consequence	ALLOW HOLD DENY ESCALATE	Execution	Outcome / new reality

Worked building example

Scenario: An ML model predicts that an air-handling unit is approaching a condition that could materially degrade performance. The model recommends an immediate control change.

01 REALITY	Equipment and occupied space are in a real physical state.
02 RECORD	Sensors and system records capture observations attributable to source, time, location, and context.
03 COMPUTATION	The ML process uses permitted evidence to infer degradation and forecast likely performance.
04 PROPOSITION	The system proposes a control change. This is not yet permission to execute.
05 ADMISSIBILITY	Is the evidence sufficient and fit for this particular decision? Are freshness, continuity, quality, context, and required corroboration satisfied?
06 AUTHORITY + STANDING	Is this control change within the established authority of this model, system, operator, or role, for this equipment, under these conditions, now?
07 COMMIT / EXECUTION	Only after the boundary conditions are satisfied may the authorized consequence cross into execution.
08 OUTCOME	Measure again. Record what actually happened. Compare outcome with intended consequence. Begin again from reality.

Human approval is not a substitute for governance.

A human click does not cure insufficient evidence, missing authority, or absent standing. The boundary governs the consequence whether the proposition originated with a human, an ML model, or both.

Practitioner vocabulary + presentation language

TERM	WORKING MEANING
Admissible Evidence	Evidence fit to support the specific proposed decision or consequence under the applicable rules and conditions.
Applicable Authority	The rule, role, permission, mandate, contract, policy, control scope, or other valid source establishing who or what may cause the consequence.
Established Standing	The demonstrated relationship of the actor or system to the specific decision: why this actor or system is entitled to exercise that authority here and now.
Proposed Consequence	A recommendation, command, intervention, transaction, disclosure, change, or other proposed effect that has not yet crossed into reality.
Commit Boundary	The last governed point at which the proposed consequence can be allowed, held, denied, or escalated before execution.
Verified Outcome	Evidence of what actually happened after execution, returned to the record rather than assumed from the command.

For presentation use

Prediction or inference does not create authority to act.

A building ML system may produce a technically valid and highly reliable inference. That inference is still a proposition when it recommends a real-world change. Before the recommendation becomes consequence, the system must establish that the evidence is admissible for that decision, that applicable authority exists, and that the actor or system has standing to exercise it now.

TA-14 architecture context

For governed building intelligence, TA-14 separates the record and computation side from the execution side:

REALITY -> AIR -> ACA -> INTELLIGENCE -> PROPOSED CONSEQUENCE -> AEA -> EXECUTION -> OUTCOME -> NEW REALITY

AIR protects what was observed and recorded. **ACA** governs what intelligence may rely upon before computation. **AEA** governs what intelligence may cause before execution. After execution, the architecture returns to reality and records the actual outcome.

Capability != Authority. Understanding does not create permission. Commissioned once does not mean authorized forever.

This reference is an architectural clarification, not legal advice and not a claim of governmental, standards-body, or third-party adoption.